

Emergent Social Behaviour & Dilemmas in Multi-Agent Reinforcement Learning

By

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Abstract

Multi-agent reinforcement learning (MARL) increasingly powers autonomous systems, from traffic control to distributed energy grids, yet a fundamental question remains open: when does an explicit *coordination mechanism* actually help, and when do independent learners suffice? This dissertation answers that question on a controlled, inherently cooperative benchmark — traffic signal control on a 2×2 grid of four intersections in SUMO, integrated with Ray RLlib.

A Multi-Agent Proximal Policy Optimization (MAPPO) controller, with a centralized critic and a neighbour-coupled reward, was implemented and trained to stable convergence. Under a *fair, matched-clearance* evaluation — in which every controller pays the same realistic safety clearance — the learned controller decisively outperforms both the fixed-time and max-pressure heuristics; a methodological finding of this work is that the heuristics’ apparent strength was an artefact of an unfair zero-clearance evaluation. The project’s MAPPO was further confirmed to match the canonical published recipe of Yu et al. (2021), establishing it as a faithful, competitive implementation rather than a weak strawman.

The central result is a surprise. Compared against Independent PPO (IPPO) — a fully decentralized comparator differing only in its decentralized critic and purely local reward — MAPPO proves *statistically indistinguishable* on every traffic-performance metric across five seeds; the only reliable difference is that the cooperative controller performs slightly more phase switches, with no performance benefit. On a 2×2 network the *coordination value* (MAPPO – IPPO) is therefore approximately zero. The interpretation is not that coordination is worthless but that the network is small enough that incentives are already locally aligned, so independent learners recover coordinated behaviour implicitly. This locates precisely where explicit coordination should begin to pay off — at larger network scales — and motivates the future work of scaling to a 5×5 grid and of extending the study to genuine social dilemmas, both placed beyond this submission by compute constraints. The contribution is thus a careful negative result together with a sharpened hypothesis: the value of cooperation in MARL is not a fixed property of an algorithm but a function of scale.

Contents

Abstract	i
1 Introduction	1
1.1 Problem Statement	1
1.2 Research Gap	1
2 Background and Literature Review	3
2.1 Reinforcement Learning	3
2.2 Deep Reinforcement Learning	3
2.3 Multi-Agent Reinforcement Learning	4
2.3.1 Fundamental Challenges	4
2.3.2 Algorithmic Approaches	4
2.4 Social Dilemmas in Multi-Agent Systems	5
2.5 MARL for Traffic Signal Control	5
3 Objectives	7
4 Methodology	8
4.1 Environment	8
4.2 Observation, Action and Reward	8
4.2.1 Observation space (70 dimensions per agent)	8
4.2.2 Action space	9
4.2.3 Reward function	9
4.3 Network Architectures	9
4.4 Algorithms Under Comparison	9
4.4.1 Paper-baseline (fidelity check)	10
4.5 Baseline Controllers	10
4.6 Fair Evaluation Protocol	10
5 Results and Analysis	12

5.1	Training Convergence	12
5.2	Comparison Against Heuristic Baselines (Fair Clearance)	13
5.3	Fidelity Check Against the Published Recipe	14
5.4	The Central Result: Coordination Value ≈ 0	14
6	Discussion	16
6.1	Why the Coordination Value Is Near Zero	16
6.2	The Scale Hypothesis	17
6.3	Implications	17
7	Future Work	18
7.1	Network Scale-Up ($2 \times 2 \rightarrow 5 \times 5$)	18
7.2	Extending to a Social Dilemma	18
7.3	Efficiency–Equity Analysis	19
8	Conclusion	20

List of Figures

5.1	MAPPO (v2) mean episode reward over training iterations. Reward rises from ≈ -2757 to a stable plateau near -50 within roughly 100 iterations.	12
5.2	Training stability. (a) KL divergence stays small, confirming PPO clipping constrains the updates. (b) The centralized critic’s explained variance climbs to ≈ 0.91 , indicating accurate value prediction.	13
5.3	Left: average waiting time for all controllers under matched 3 s clearance — the learned controllers clearly win. Right: the confound — charging the heuristics the same clearance they had been spared erases their apparent edge, lifting them well above the learned-controller band (shaded).	14
5.4	The central finding. Across all five metrics, MAPPO and IPPO are statistically indistinguishable (“n.s.”), except for a small, performance-neutral difference in phase-switch count. The explicit coordination machinery buys no measurable advantage on the 2×2 network.	15

List of Tables

2.1	Comparison of representative MARL algorithms.	5
4.1	Shared training hyperparameters (MAPPO v2 and IPPO).	10
5.1	Average waiting time (s) by controller, with and without the 3 s safety clearance (mean $\pm$ std over seeds 42–46). The learned controllers were always evaluated at 3 s; only the matched-clearance column is a fair comparison.	13
5.2	MAPPO versus IPPO across all evaluation metrics (mean $\pm$ std over seeds 42–46; p from Welch’s t -test). Only the phase-switch count differs significantly, and it favours neither controller in performance terms.	15

Chapter 1

Introduction

1.1 Problem Statement

Multi-agent systems — where several autonomous agents interact to achieve goals individually or collectively — underpin a growing share of modern technology. Autonomous vehicles share roads, distributed sensors coordinate surveillance, and energy grids balance supply across many producers. Unlike traditional control systems with a single centralized decision maker, these settings contain independent agents that must learn to coexist, coordinate, or compete. The difficulty is not merely making each agent individually competent; it is understanding what emerges when many *learning* agents interact.

Reinforcement learning (RL) is a paradigm in which an agent learns to act by interacting with an environment and receiving reward feedback. Extending RL to many simultaneously learning agents — Multi-Agent Reinforcement Learning (MARL) — changes the problem fundamentally. Each agent treats the others as part of its environment, yet those “parts” are themselves learning and changing: today’s optimal response can fail tomorrow once the others adapt. This *non-stationarity* breaks the assumptions that make single-agent RL convergent and stable.

Crucially, the *relationship* between agents varies. Sometimes cooperation benefits everyone; at other times genuine conflict arises, and an agent can exploit a shared resource for private gain while pushing the cost onto the collective. Knowing *when* cooperation emerges versus *when* exploitation dominates is central to deploying autonomous agents responsibly.

This dissertation studies cooperation in MARL using *traffic signal control* as a representative, inherently cooperative case study. Each intersection is an agent; because network effects mean that gridlock hurts everyone (including the intersection that caused it), traffic sits at the *pure-coordination* end of the cooperation spectrum and is an ideal testbed for isolating the value of explicit coordination mechanisms.

1.2 Research Gap

Despite rapid progress, several questions remain underexplored:

- **The middle of the spectrum.** Most work targets either purely cooperative settings

or purely competitive games (chess, poker). The continuum between them — where coordination and competition coexist — receives less systematic study.

- **Reward structure.** While reward shaping is known to matter, controlled comparisons of *individual* versus *shared* objectives across standardized environments are rare.
- **Fairness.** Algorithms typically optimize total system performance without examining how benefits distribute across agents.
- **Scalability of coordination.** How the *value* of a coordination mechanism changes as a system grows — from a handful of agents to large networks — has not been systematically characterized.

This work contributes to the first and fourth gaps directly. It asks a deceptively simple question on a controlled traffic benchmark: *does an explicitly cooperative algorithm (MAPPO, with a centralized critic) actually outperform a set of independent learners (IPPO) when the two are matched in every other respect?* The answer — and the reason behind it — forms the spine of this dissertation.

Chapter 2

Background and Literature Review

2.1 Reinforcement Learning

Reinforcement learning provides a mathematical framework for an agent learning optimal behaviour through interaction. Its foundation is the Markov Decision Process (MDP): a set of *states* describing the environment, *actions* the agent may take, *transition dynamics* mapping state–action pairs to next states, and a *reward* signal. The agent seeks a *policy* — a mapping from states to actions — that maximizes the expected cumulative discounted return. This is captured by the Bellman equation [1], which expresses the value of a state as the immediate reward plus the discounted value of its successor, enabling algorithms to *bootstrap* value estimates from one another. Classical methods such as Q-learning iteratively estimate the optimal action-value function after each interaction, addressing the *credit assignment* problem of attributing outcomes to past actions.

2.2 Deep Reinforcement Learning

Tabular methods store a value for every state–action pair and become infeasible in large or continuous state spaces. Deep Reinforcement Learning (DRL) replaces the table with a neural-network function approximator. The Deep Q-Network (DQN) pioneered this with *experience replay* (sampling past transitions to break temporal correlation) and *target networks* (stabilizing the bootstrap target).

Whereas DQN is value-based, *policy-gradient* methods directly optimize the policy by ascending the expected return. Proximal Policy Optimization (PPO) [2] has become the standard modern choice: it constrains how far the policy may move in a single update through a clipped surrogate objective, preventing the destructive updates that destabilized earlier methods while remaining simple and sample-efficient. Generalized Advantage Estimation (GAE) [3] complements PPO by trading off bias and variance in the advantage estimate through exponentially weighted temporal-difference errors; it is now standard in policy-gradient implementations.

2.3 Multi-Agent Reinforcement Learning

2.3.1 Fundamental Challenges

Extending RL to multiple agents introduces complications absent in the single-agent case. **Non-stationarity** is the central one: each agent perceives the others as part of its environment, but those components evolve as the others learn, violating the stationary-transition (Markov) assumption and voiding single-agent convergence guarantees. **Credit assignment** also becomes harder: when a team shares a reward, attributing it to individual actions is difficult, and the joint action space grows exponentially (e.g. four agents with four actions yield $4^4 = 256$ combinations). **Scalability** therefore demands decomposing the global problem into local subproblems each agent can solve from its own observation.

2.3.2 Algorithmic Approaches

Independent Q-Learning (IQL) [4] is the simplest approach: each agent learns its own value function, treating the others as environment and ignoring non-stationarity entirely. It is simple and scalable, often works in practice when mutual influence is limited, but lacks convergence guarantees and struggles with cooperative credit assignment. Its policy-gradient analogue, **Independent PPO (IPPO)**, applies PPO independently per agent and is the “fully independent learners” baseline used throughout this work.

Value-decomposition methods address credit assignment by learning how individual values combine into a team value. **QMIX** [5] represents the joint action-value as a *monotonic* mixture of individual values, guaranteeing that improving an individual’s value improves the team’s, while permitting decentralized execution.

These advances rest on the **Centralized Training with Decentralized Execution (CTDE)** paradigm. Under CTDE, training may use privileged global information (all agents’ observations and actions) to stabilize learning and tame non-stationarity, but the deployed policies act only on local observations, preserving scalability and requiring no inter-agent communication at run time. A useful analogy is a sports team: in practice a coach observes all players and teaches coordination; in the match, players act independently on their local view, applying what they learned.

MADDPG [6] pioneered CTDE for actor–critic methods with centralized critics that see all agents’ observations and actions during training and decentralized actors at execution. **MAPPO** [7] combines this centralized-critic architecture with PPO’s robust optimization: a shared centralized value function consumes all agents’ observations during training, each agent keeps its own decentralized policy, PPO clipping prevents destructive updates, and GAE stabilizes advantages. MAPPO is reported to match or beat more specialized cooperative algorithms across standard benchmarks, which is precisely why it is the natural “sophisticated cooperative method” to test here.

Table 2.1 summarizes these algorithms.

Table 2.1: Comparison of representative MARL algorithms.

Algorithm	Training	Execution	Key characteristics
IQL / IPPO	Independent	Decentralized	Simple, scalable; treats others as environment; unstable under non-stationarity
QMIX	Centralized mixer	Decentralized	Monotonic value factorisation; cooperative settings only
MADDPG	Centralized critic	Decentralized	Actor-critic CTDE; continuous actions; less stable, limited scalability
MAPPO	Centralized critic	Decentralized	PPO-based; stable; strong coordination and scalability

2.4 Social Dilemmas in Multi-Agent Systems

Social dilemmas describe situations where individually rational behaviour yields a collectively poor outcome. The *Prisoner’s Dilemma* is canonical: both agents benefit from mutual cooperation, yet each maximizes its own payoff by defecting regardless of the other, so rational self-interest drives both to a worse outcome. The *Tragedy of the Commons* extends this to shared resources: each user benefits from exploitation while the cost is distributed, creating pressure toward overexploitation that destroys the resource for all. *Sequential social dilemmas* [8] embed such incentives in grid-world environments over many timesteps; the “Harvest” commons, where agents collect slowly regrowing apples and over-harvesting permanently depletes a region, is a standard example. Social dilemmas sit at the *opposite* end of the spectrum from traffic and are discussed as future work (Chapter 7).

2.5 MARL for Traffic Signal Control

Traffic signal control is naturally modelled as MARL: each traffic light is an agent that observes local traffic state, selects a signal phase, and is rewarded for traffic efficiency. Because intersections interact through traffic flow, agents must learn policies that are good for the whole network, not just locally. Traditional fixed-time and actuated controllers require manual tuning and cope poorly with non-recurring congestion; RL promises adaptive controllers that learn from experience. Early single-intersection deep-RL controllers improved on classical methods but missed network effects, where a locally optimal decision creates downstream problems.

CoLight [9] addressed this by giving each agent information about adjacent intersections and sharing network parameters across agents, inducing coordinated behaviour despite decentralized execution. A particularly effective family is *pressure-based* control, which minimizes the imbalance between incoming and outgoing vehicle counts per movement and connects to max-pressure control from transportation theory. A large-scale comparison [10] found that sharing neighbour information significantly improves network performance *but only with careful reward design*: naive individual rewards, where each intersection minimizes only its own delay, can harm overall throughput, whereas shared

rewards help but demand careful algorithm design to scale.

A central observation motivates this dissertation: from a system designer’s viewpoint traffic is *fundamentally cooperative*. Network effects align incentives — gridlock hurts everyone — which makes traffic an excellent, controlled environment for asking whether an explicit coordination mechanism is actually *necessary*, or whether independent learners can recover the same behaviour on their own.

Chapter 3

Objectives

The overarching aim of this research is to study emergent social behaviour, cooperation, and the value of coordination in Multi-Agent Reinforcement Learning, using traffic signal control as a controlled, inherently cooperative case study. The realized objectives of this dissertation are:

1. **Establish a strong cooperative baseline.** Implement and train a MAPPO traffic-signal controller on a 2×2 grid and demonstrate stable convergence, then validate it against the standard heuristic controllers (fixed-time and max-pressure) under a *fair* evaluation protocol.
2. **Confirm implementation fidelity.** Compare the project’s MAPPO configuration against the canonical published recipe of Yu et al. [7], to establish that the cooperative agent is a faithful, competitive implementation and not a weakened strawman.
3. **Quantify the value of coordination.** Implement Independent PPO (IPPO) as a fully decentralized comparator that differs from MAPPO only in its two defining design choices — a decentralized critic and a purely local reward — and measure the *coordination value*, defined as the performance difference MAPPO – IPPO.
4. **Characterize and interpret the result.** Analyze *why* the measured coordination value takes the value it does at this network scale, and connect the finding to the broader question of how coordination mechanisms scale with network complexity.

Two further aims — characterizing the efficiency–equity trade-off and contrasting coordination against genuine social dilemmas — motivate the future work outlined in Chapter 7, where network scale-up and a social-dilemma environment are identified as the natural next steps that compute constraints placed beyond the scope of this submission.

Chapter 4

Methodology

This chapter describes the environment, the agent design shared by all learned controllers, the two algorithms under comparison (MAPPO and IPPO), the published paper-baseline used as a fidelity check, and — importantly — the *fair* evaluation protocol under which all controllers are compared.

4.1 Environment

The environment is a 2×2 grid network of four signalized intersections (J1–J4), each controlled by one agent, simulated in SUMO (Simulation of Urban MObility) and driven through the TraCI interface, integrated with the Ray RLlib framework. Roads between intersections are 150 m and entry/exit roads 100 m; each road has three lanes with strict directional assignment (lane 0 right turns, lane 1 straight, lane 2 left turns). Lane-area detectors span each lane for full observability of positions, speeds and waiting times. An episode lasts 3600 s of simulation with a 5 s action interval, giving up to 720 decision points per agent. Demand follows a realistic profile: light traffic (0–600 s), a rush hour (600–2400 s), then light traffic again.

4.2 Observation, Action and Reward

4.2.1 Observation space (70 dimensions per agent)

Each agent observes a 70-dimensional vector: **28 local** features — per-movement queue lengths (12), one-hot current phase direction (2), normalized elapsed phase time (1), movement pressures, i.e. incoming minus outgoing counts (6), pressure derivatives (6), and a binary min-green flag (1) — and **42 neighbour** features from the two adjacent intersections (21 each), carrying shared queues, neighbour phase, combined pressures, and directional incoming/outgoing metrics, spatially discounted with $\gamma_s = 0.9$ to weight nearer intersections more heavily.

4.2.2 Action space

The action space is discrete with four green phases following traffic-engineering standards: north–south through/right, north–south left, east–west through/right, and east–west left. Right turns are permissive in all phases.

4.2.3 Reward function

The reward combines five weighted components (configuration v2):

$$r_t = -1.0 W_t - 0.25 Q_t + 0.1 T_t - 0.4 P_t - 0.4 N_t, \quad (4.1)$$

where W_t is cumulative waiting time (the primary objective), Q_t the queue length, T_t the network throughput (a shared term), P_t the local pressure (incoming minus outgoing queue imbalance), and N_t the spatially discounted *neighbour* pressure ($\gamma_s = 0.9$) that couples each agent to its neighbours. The reward is clipped to $[-3.0, 1.0]$. The neighbour term N_t is the sole explicit multi-agent coupling in the reward and is the lever that distinguishes MAPPO from IPPO (Section 4.4).

4.3 Network Architectures

All learned agents share a single policy (parameter sharing across the four homogeneous intersections). The **actor** is decentralized: it maps the agent’s 70-dimensional local observation through two hidden layers of 256 units with tanh activations to four action logits. The MAPPO **critic** is *centralized*: it consumes the 280-dimensional global state (the four agents’ 70-dimensional observations concatenated) through hidden layers of 512, 256 and 128 units with ReLU activations to a single value. Orthogonal initialization and a running observation normalizer (`MeanStdFilter`) are used throughout.

4.4 Algorithms Under Comparison

The comparison is designed so that MAPPO and IPPO differ *only* in the two design choices that define “cooperative MARL” versus “independent learners”; every other hyperparameter and the actor architecture are identical.

- **MAPPO (cooperative).** Centralized critic over the 280-dimensional global state; the neighbour-pressure reward term N_t is active. This is the “full cooperative package”: CTDE training plus an explicitly coordination-aware reward.
- **IPPO (independent).** Decentralized critic that sees only the agent’s own 70-dimensional observation; the neighbour-pressure weight is set to zero so each agent optimizes a purely local objective. Parameter sharing is retained. This is the “fully independent learners” package.

The shared hyperparameters (configuration v2) are listed in Table 4.1.

Table 4.1: Shared training hyperparameters (MAPPO v2 and IPPO).

Hyperparameter	Value	Note
Learning rate	5×10^{-4}	
Discount factor γ	0.99	long-horizon optimization
GAE λ	0.95	bias-variance trade-off
PPO clip ϵ	0.2	
Value clip	10.0	
Gradient clip	1.0	
Entropy coefficient	0.02	exploration
Value-loss coefficient	1.0	
SGD iterations/update	15	
Minibatch / train batch	32768/32768	full-batch update
Rollout workers	3	
Rollout fragment	200	complete-episode batching
Training iterations	200	
Observation filter	MeanStdFilter	running mean/std

4.4.1 Paper-baseline (fidelity check)

To confirm that the project’s MAPPO is not an under-powered strawman, it is also compared against the canonical recipe of Yu et al. [7] (their adopted MLP preset): actor and critic both two hidden layers of 512 units with ReLU, actor learning rate 7×10^{-4} , 15 SGD iterations per update, a single full-batch minibatch, clip 0.2, entropy coefficient 0.015, and gradient clipping at 10.0. All three controllers (MAPPO v2, IPPO, paper-baseline) are trained for 200 iterations under identical environment dynamics.

4.5 Baseline Controllers

Two standard heuristics provide reference points. The **fixed-time** controller cycles phases on a fixed schedule. The **max-pressure** controller selects, at each decision, the phase that maximizes the relief of pressure (incoming minus outgoing queue imbalance), subject to a minimum green time.

4.6 Fair Evaluation Protocol

A controlled comparison requires that every controller pay the same real-world operating costs. Two points are essential.

Matched safety clearance. The learned agents operate with a 3s all-red clearance on every phase change (`min_red = 3`), a realistic safety constraint they were *trained* under. The heuristic controllers, by default, switched phases instantaneously (zero clearance), which silently advantaged them. All controllers are therefore evaluated with the *same* 3s clearance; the consequences of ignoring this are quantified in Chapter 5.

Faithful policy evaluation. With RLlib’s connector-enabled inference, the running observation normalizer is not applied automatically at evaluation; it is applied manually so the deterministic policy receives the same input distribution it saw in training (omitting it collapses the policy). Neighbour edge-connectivity metadata is likewise forwarded into the evaluation environment so the full 70-dimensional observation is reconstructed. Each controller is evaluated over five seeds (42–46); per-metric means, standard deviations, and Welch’s t -tests are reported in Chapter 5.

Chapter 5

Results and Analysis

This chapter presents the empirical findings in the order that tells the story: the cooperative controller is first shown to train well and to beat the heuristic baselines under a fair evaluation; it is then confirmed to be a faithful implementation of the published method; and finally it is compared head-to-head against independent learners, with a result that inverts the natural expectation.

5.1 Training Convergence

The MAPPO controller (configuration v2) trains stably to convergence over 200 iterations. The mean episode reward improves from -2756.5 at initialization to a converged last-ten-iteration mean of -49.76 ± 0.41 (Figure 5.1). The centralized critic is accurate, reaching a value-function explained variance of 0.907, and policy updates remain conservative, with an average KL divergence of 0.0034 across the final iterations (Figure 5.2). Together these indicate healthy, stable joint optimization of the actor and critic without divergence or value collapse.

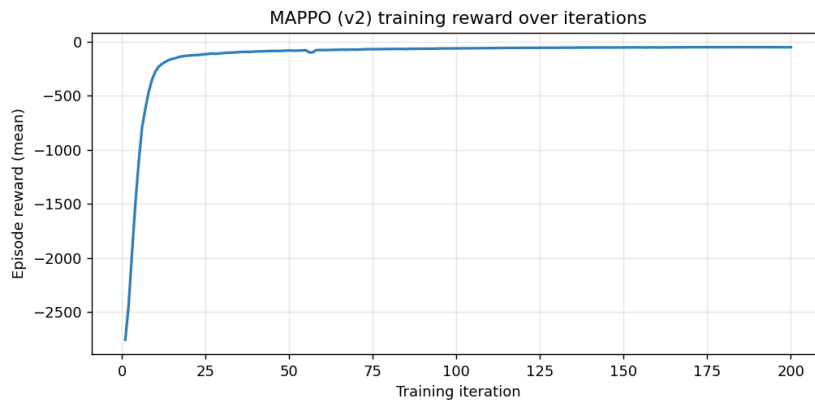


Figure 5.1: MAPPO (v2) mean episode reward over training iterations. Reward rises from ≈ -2757 to a stable plateau near -50 within roughly 100 iterations.

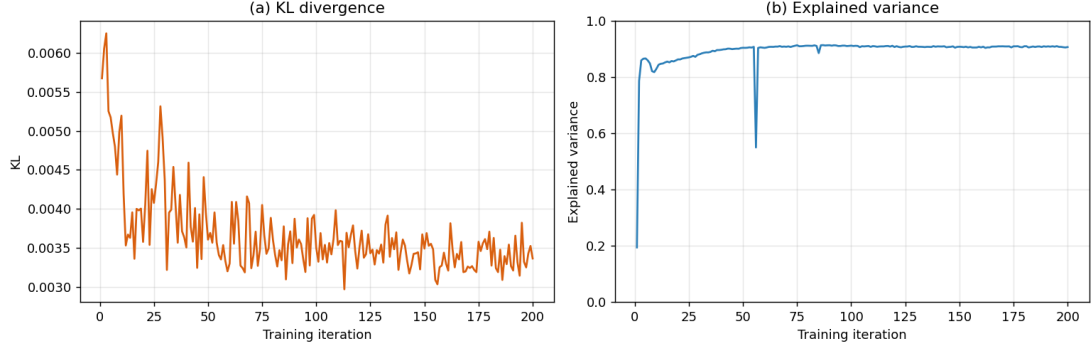


Figure 5.2: Training stability. (a) KL divergence stays small, confirming PPO clipping constrains the updates. (b) The centralized critic’s explained variance climbs to ≈ 0.91 , indicating accurate value prediction.

5.2 Comparison Against Heuristic Baselines (Fair Clearance)

A subtle but decisive evaluation issue must be addressed first. The learned agents pay a 3s all-red safety clearance on every phase change, whereas the heuristic controllers, by default, switched instantaneously. Evaluated under that *unfair* zero-clearance setting, max-pressure appears competitive (8.15 ± 0.08 s average wait). But when the *same* 3s clearance is applied to the heuristics — the cost the learned agents always paid — max-pressure’s average wait nearly triples to 21.63 ± 1.00 s and fixed-time rises to 47.03 ± 0.59 s (Table 5.1, Figure 5.3). Under this fair, matched-clearance protocol the MAPPO controller (9.92 ± 0.65 s) outperforms max-pressure by roughly a factor of two and fixed-time by nearly a factor of five. The heuristics’ apparent strength was therefore an artefact of an unequal evaluation, not a genuine advantage; correcting it both restores a fair comparison and strengthens the case for the learned controllers.

Table 5.1: Average waiting time (s) by controller, with and without the 3s safety clearance (mean $\pm$ std over seeds 42–46). The learned controllers were always evaluated at 3s; only the matched-clearance column is a fair comparison.

Controller	0 s clearance (unfair)	3 s clearance (fair)
MAPPO (v2)	—	9.92 ± 0.65
IPPO	—	10.34 ± 0.31
Max-Pressure	8.15 ± 0.08	21.63 ± 1.00
Fixed-Time	38.02 ± 0.81	47.03 ± 0.59

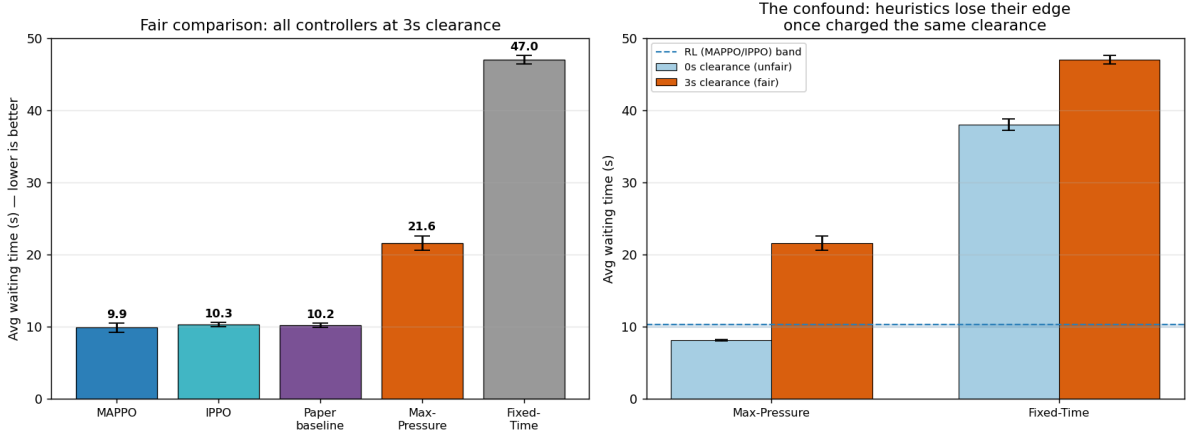


Figure 5.3: Left: average waiting time for all controllers under matched 3s clearance — the learned controllers clearly win. Right: the confound — charging the heuristics the same clearance they had been spared erases their apparent edge, lifting them well above the learned-controller band (shaded).

5.3 Fidelity Check Against the Published Recipe

To rule out the possibility that the cooperative controller is an under-powered strawman, it is compared against the canonical MAPPO recipe of Yu et al. [7] (512×512 networks; see Chapter 4). The two are statistically indistinguishable on average wait (9.92 ± 0.65 s for the project’s MAPPO versus 10.21 ± 0.28 s for the paper recipe; Welch’s t -test $p = 0.39$), with the project’s configuration numerically slightly ahead. The implemented MAPPO is therefore a faithful and competitive representative of the method — which makes the comparison that follows meaningful: any failure to beat independent learners cannot be blamed on a weak MAPPO.

5.4 The Central Result: Coordination Value ≈ 0

MAPPO is the sophisticated, theoretically motivated choice, with a centralized critic designed to tame non-stationarity and solve credit assignment and a neighbour-coupled reward that explicitly encodes coordination. One expects it to beat a set of independent learners that ignore all of this. It does not.

Against IPPO — identical in every respect except its decentralized critic and its purely local reward — MAPPO is *statistically indistinguishable on every traffic-performance metric* across five seeds (Table 5.2, Figure 5.4): average waiting time ($p = 0.24$), average halting ($p = 0.53$), maximum halting ($p = 0.74$), and completed trips ($p = 0.55$) all show no significant difference. The *only* reliable difference is that MAPPO performs about 21 more phase switches per episode (1391 ± 12 versus 1369.6 ± 8.4 , $p = 0.015$) — additional control activity that yields no measurable performance benefit. The coordination value, defined as MAPPO – IPPO, is thus approximately zero on this network.

Table 5.2: MAPPO versus IPPO across all evaluation metrics (mean $\pm$ std over seeds 42–46; p from Welch’s t -test). Only the phase-switch count differs significantly, and it favours neither controller in performance terms.

Metric	MAPPO	IPPO	p -value
Average wait (s)	9.92 ± 0.65	10.34 ± 0.31	0.236
Average halting	3.91 ± 0.11	4.00 ± 0.27	0.534
Maximum halting	14.2 ± 1.6	14.6 ± 2.1	0.745
Completed trips	1261.6 ± 0.9	1261.2 ± 1.1	0.545
Phase switches	1391 ± 12	1370 ± 8	0.015

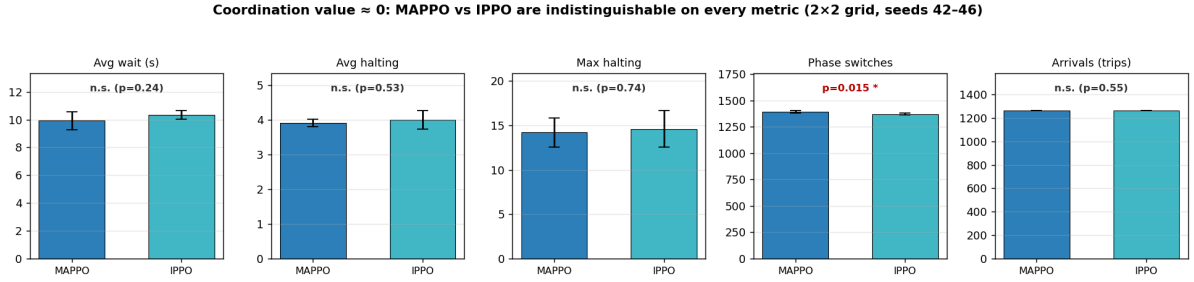


Figure 5.4: The central finding. Across all five metrics, MAPPO and IPPO are statistically indistinguishable (“n.s.”), except for a small, performance-neutral difference in phase-switch count. The explicit coordination machinery buys no measurable advantage on the 2 $\times$ 2 network.

This is the plot twist the dissertation set out to surface: on a small, inherently cooperative network, the full cooperative MARL package performs no better than independent learners. Chapter 6 examines why, and Chapter 7 sets out the experiments that would test the resulting hypothesis.

Chapter 6

Discussion

The results of Chapter 5 invert the expectation that frames most of the cooperative-MARL literature. MAPPO is the sophisticated, theoretically motivated choice: its centralized critic is designed precisely to tame non-stationarity and to solve the multi-agent credit-assignment problem, and its neighbour-coupled reward explicitly encodes coordination. One naturally expects it to beat a set of independent learners that ignore all of this. Yet on the 2×2 network the two are statistically indistinguishable on every performance metric, and the only reliable difference — MAPPO’s slightly higher switching rate — is, if anything, a mild inefficiency rather than a benefit. This chapter asks why.

6.1 Why the Coordination Value Is Near Zero

The most parsimonious explanation is that, at this scale, *there is little coordination left for an explicit mechanism to add*. A 2×2 grid has only four intersections, each adjacent to two neighbours, and the traffic-control reward is already locally informative: minimizing one’s own queue and waiting time, on a network this tightly coupled, is almost the same objective as minimizing the network’s. In other words, the incentives are *already aligned* by the structure of the problem. The centralized critic provides a lower-variance value estimate and the neighbour-pressure term nudges toward network-aware behaviour, but an independent learner equipped with the same local observations and the same parameter-sharing can discover the same coordinated phasing on its own. Coordination here is *emergent* rather than *engineered*: it falls out of independent optimization because the environment does not reward defection.

This reading is reinforced by two details. First, both learned controllers *decisively* beat the heuristic baselines under matched clearance (Chapter 5); the learning is doing real work — it is not that the task is trivial, but that the *coordination-specific* additions are redundant. Second, MAPPO’s extra phase switches buy no improvement, which is consistent with a centralized critic that is active but not pivotal: it shapes the policy slightly differently without changing what the policy ultimately achieves.

6.2 The Scale Hypothesis

If the coordination value is small *because* the network is small, then it should grow with scale. As a grid enlarges, an agent’s actions influence ever more distant intersections through longer chains of traffic; the gap between “minimize my own delay” and “minimize the network’s delay” widens, non-stationarity from the perspective of any single learner intensifies, and the centralized critic’s access to the global state should translate into a measurable advantage. This is exactly the regime in which network-level cooperation has been shown to matter: CoLight [9] demonstrates gains from sharing neighbour information at city scale, and the large-scale study of Chu et al. [10] reports that neighbour-aware, carefully designed shared rewards improve performance — while naive individual rewards can actively harm throughput — on large networks. The near-zero coordination value measured here is therefore not in tension with that literature; it is the expected *small-scale* end of the same curve.

6.3 Implications

The practical implication is a note of caution and a research direction. The caution is that, on small or tightly coupled multi-agent problems, the considerable machinery of CTDE may be unnecessary: a simpler independent-learning approach with parameter sharing can match it, at lower implementation and training cost. The research direction is that the *value of coordination* should be treated as a quantity that depends on system scale and on how strongly incentives conflict — not as a fixed property of an algorithm. The controlled traffic setting used here establishes a clean zero-point for that quantity, against which both larger networks and genuine social dilemmas (Chapter 7) can be measured.

Chapter 7

Future Work

The central finding of this dissertation — that an explicit coordination mechanism buys essentially nothing over independent learners on a 2×2 traffic network — is best read not as an endpoint but as a precisely located starting point. Two directions follow directly, both placed beyond the scope of this submission by compute constraints rather than by design.

7.1 Network Scale-Up ($2 \times 2 \rightarrow 5 \times 5$)

The most direct test of the interpretation in Chapter 6 is to grow the network. If the coordination value is near zero because a 2×2 grid is small enough that incentives are already locally aligned, then enlarging the network to 5×5 (25 intersections) should widen the gap between MAPPO and IPPO: distant intersections interact through longer traffic chains, non-stationarity intensifies, and the centralized critic’s global view should begin to pay off. This expectation is consistent with the large-scale findings of Chu et al. [10] and the network-level cooperation of CoLight [9]. A 5×5 SUMO network skeleton was prototyped during this project but reverted; training it to convergence at this scale (with a single GPU and per-run training times measured in days) exceeded the available compute budget. Quantifying how coordination value scales with network size is therefore the primary item of future work.

7.2 Extending to a Social Dilemma

Traffic occupies the pure-coordination end of the cooperation spectrum: incentives are aligned, so there is nothing to defect over. The complementary experiment is to move to the *opposite* end — a sequential social dilemma in which defection is individually tempting but collectively harmful. A natural choice is an apple-harvesting commons in the style of Leibo et al. [8], where agents collect a slowly regrowing resource and over-harvesting permanently depletes it. There, the reward-structure axis (individual versus shared rewards) that is inert in traffic should become decisive, allowing a direct study of when shared rewards are *necessary* to sustain cooperation versus when independent learning suffices. Contrasting the learned behaviours across the two environments would complete the coordination–dilemma spectrum this project set out to chart.

7.3 Efficiency–Equity Analysis

Finally, the per-agent breakdowns collected here can support a fairness analysis (e.g. a Gini coefficient over per-intersection outcomes) to determine whether high system performance requires unequal treatment of individual intersections — a question that becomes more pointed at larger scales and in dilemma settings.

Chapter 8

Conclusion

This dissertation set out to test a simple but consequential question on a controlled traffic benchmark: does an explicitly cooperative multi-agent algorithm actually outperform a set of independent learners when the two are matched in every other respect? The investigation produced a clear, and initially surprising, answer.

A MAPPO traffic-signal controller was implemented on a 2×2 grid, trained to stable convergence, and shown — under a *fair*, matched-clearance evaluation — to outperform both the fixed-time and max-pressure heuristics decisively. A methodological contribution emerged along the way: the heuristic baselines had been flattered by an unfair zero-clearance evaluation, and correcting it (charging every controller the same safety clearance) both restored a fair comparison and strengthened the case for the learned controllers. The project’s MAPPO was further confirmed to be a faithful implementation by matching the canonical published recipe.

The central result is that, against this strong and faithful MAPPO, an Independent PPO comparator — differing only in its decentralized critic and purely local reward — proved *statistically indistinguishable* on every traffic-performance metric. The sophisticated coordination machinery bought no measurable advantage; its only distinguishing behaviour was a slightly higher rate of phase switching that produced no benefit. On a 2×2 network, the *coordination value* is approximately zero.

The interpretation is not that coordination is worthless, but that the testbed is too small to require it: at this scale incentives are already locally aligned, and independent learners recover coordinated behaviour implicitly. This precisely locates where explicit coordination should begin to matter — at larger network scales — and motivates the future work of scaling to 5×5 and of extending the study to genuine social dilemmas, where reward structure and the temptation to defect are expected to make coordination indispensable. The contribution of this work is thus both a careful negative result and a sharpened hypothesis: the value of cooperation in multi-agent reinforcement learning is not a constant but a function of scale, and the controlled traffic setting establishes the baseline against which that function can now be measured.

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